

The plumbotectonic model for Pb isotopic systematics among major terrestrial reservoirs— A case for bi-directional transport*

ROBERT E. ZARTMAN and SARA M. HAINES

U.S. Geological Survey, Mailstop 963, Denver Federal Center, Denver, CO 80225, U.S.A.

(Received August 13, 1987; accepted in revised form March 7, 1988)

Abstract—Version IV of plumbotectonics expands and refines the original model of DOE and ZARTMAN (1979) and ZARTMAN and DOE (1981) for explaining Pb (Sr, and Nd) isotopic systematics among major terrestrial reservoirs. A case for bi-directional transport among reservoirs is based on the observed isotopic compositions for different tectonic settings, and finds a rationale in the kinetics of plate tectonics. Chemical fractionation and radioactive decay create isotopic differences during periods of isolation of one reservoir from another, whereas dynamic processes allowing mixing between reservoirs tend to reduce these differences. Observed isotopic characteristics reflect a balance between these opposing tendencies and provide constraints on the extent and timing of fractionation and mixing processes.

Plumbotectonics does not require interaction with a lower mantle or core reservoir over most of the Earth's lifetime, and, in fact, achieves a material balance consistent with no such exchange of material.

Important evidence of the amount and timing of crustal recycling, and of the residence times of mantle heterogeneities lies in the coupled $^{207}\text{Pb}/^{204}\text{Pb}$ – $^{206}\text{Pb}/^{204}\text{Pb}$ systematics. We believe that examination of the published data base fully supports our contention of significant bi-directional transport of material among terrestrial reservoirs. Plumbotectonics allows us to explore many aspects of reservoir interaction, and to identify parameters that provide meaningful constraints on mantle-crust differentiation. We put forth a compromise fit to many of the model variables in version IV, which can serve as a reference for future work.

1. INTRODUCTION

DOE and ZARTMAN (1979) and ZARTMAN and DOE (1981) introduced the plumbotectonics model to account for variations in average lead isotopic composition among different tectonic settings. Originally used to explain the distinct lead isotopic provinces in western United States (DOE, 1967; ZARTMAN, 1974), the approach has proven applicable worldwide. Subsequently, this work was expanded to include strontium and neodymium isotopes (ZARTMAN, 1984a), although these single-parent decay schemes inherently provided little additional constraint on the model. Whereas the isotopic variation recorded in the systematics of Sm–Nd and Rb–Sr primarily reflects chemical fractionation between the crust and mantle, lead isotopes cannot be so simply explained.

Both the most and the least radiogenic lead occur in the continental crust, attesting to the strong partitioning of U, Th, and Pb in the upper and lower portions of the crust. Because the quantity of U, Th, and Pb presently contained in the crust exceeds by perhaps an order of magnitude that remaining in the mantle, these elements are especially sensitive to bidirectional transport among the major terrestrial reservoirs. Also, because of the invariance of $^{238}\text{U}/^{235}\text{U}$ at any time, the coupling of the $^{238}\text{U} \rightarrow ^{206}\text{Pb}$ and $^{235}\text{U} \rightarrow ^{207}\text{Pb}$ decay schemes offers a time constraint based solely on the lead isotopic composition and independent of absolute concentration. Emphasis throughout this paper will continue to be on the U–Th–Pb decay scheme, for which a revised version of plumbotectonics is derived. Accompanying, updated values for the Sr and Nd isotope parameters are included for completeness, but otherwise these decay schemes are discussed

only incidentally to our primary objective—the interpretation of Pb isotopic systematics among major terrestrial reservoirs.

Plumbotectonics simulates an evolving Earth, in which the resultant isotopic patterns arise from the cumulative effects of fractionation and mixing of parent and daughter nuclides. A plate tectonics framework has been adopted to provide quantitative estimates of chemical partitioning and recycling rates. This geologic basis gives rise to much of the model's vocabulary, but its underlying premise of bi-directional transport of material between the crust and mantle could be accommodated by dynamic processes other than modern plate activity. Even a fundamentally different mechanism operating during the early history of the Earth need not invalidate the model if it involved an equivalent magnitude of reservoir interaction.

Many of the model parameters (*e.g.*, mantle reservoir size, residence times, element distribution coefficients, etc.) as originally defined were poorly constrained. Because the isotopic differences arising between two fractionating reservoirs are a function of both parent-daughter ratios and elapsed time, various combinations of these two parameters were found to produce a similar, final isotopic composition. Continuous exchange between reservoirs generally reduces isotopic differences and further complicates fractionation relationships. Strong recycling can even produce a quasi-steady-state situation, whereby a virtually constant isotopic difference is maintained that is independent of reservoir age. Despite these inherent uncertainties, some important new insights into the isotopic and geochemical data base prompt us to give an updated version of the model.

The strongest evidence for communication among reservoirs arises when a reservoir's isotopic composition lies unequivocally outside the field that could be produced *in situ* and by itself. Closed systems must show isotopic evolution compatible with their chemistry, and, in the past, the parent-

* Presented at the Conference on "Isotope Tracers in Geochemistry and Geophysics" in honor of Professor Gerald J. Wasserburg's Sixtieth Birthday, March 23–25, 1987, Pasadena, California.

daughter ratio of a reservoir has sometimes been assumed on the basis of isotopic composition. We are now convinced that disparities do exist, and that these disparities demand that all major terrestrial reservoirs are essentially open systems. This decoupling of chemistry and isotopic ratios can be used to set limits on concentration levels and exchange rates among the interacting reservoirs. Thus, an additional utility of the model lies in exploring the permissible range of its variables. Beyond such permissible ranges, serious discrepancies arise between model and observed data or reasonable estimations.

We proceed with a description of the revised model, which resembles earlier constructions, although it has now become mathematically considerably more complex. The effects of varying certain parameters will then be examined in order to establish their reasonable ranges. In practice, we arrived at our present version IV of the model in somewhat the reverse order; that is, plausibility was determined after many iterations to find the best fit between model parameters and observed data. This version, too, will be undoubtedly subject to additional modification. Its purpose is to serve as a reference for the design of future experiments and against which new analytical results can be compared.

2. VERSION IV OF PLUMBOTECTONICS

Version IV of plumbotectonics adopts and expands upon the mathematical treatment and nomenclature as were defined and used for previous versions of the model (DOE and ZARTMAN, 1979; ZARTMAN and DOE, 1981; ZARTMAN, 1984a). The interaction between major reservoirs, *mantle*, *upper crust*, *lower crust*, and *subcrustal lithosphere*, occurs through cycles called orogenies of discrete, sequential operations that are expressed by a series of algebraic equations. This use of orogeny is something of a misnomer in that individual cycles need not represent geologically recognized mountain-building events, but, as will be explained, they do serve to quantify the global distribution of orogenic activity. The mathematical formulation of version IV is set forth in the Appendix. A computer program of the model written in Hewlett-Packard Series, 200/300 BASIC is available separately (HAINES and ZARTMAN, 1988).

As previously, we restrict the *mantle* to include only its upper portion, which now is defined as lying above the pronounced seismic discontinuity at 670 km depth. For reasons discussed in following sections of the paper, we shall also continue to represent this *mantle* by an average chemical and isotopic composition despite its obvious heterogeneity (ALLÈGRE *et al.*, 1987; ZINDLER and HART, 1986; WHITE, 1985; SUN, 1980; O'NIONS *et al.*, 1979; TATSUMOTO, 1978). If, as we maintain, most of the heterogeneity is destroyed on a time scale considerably less than the age of the Earth, this simplification of the *mantle* is probably justified. Nonetheless, to the extent that the extraction of the crust and its recycling into the mantle is an important determinant of mantle heterogeneity, adaption of plumbotectonics to better specify the various chemical and isotopic trends recorded in mantle-derived volcanic rocks becomes a desirable future objective.

During each orogeny, an *orogene* comprised of *proximal*, *distal*, and *mantle wedge* components is created and receives contributions from the *mantle* and from preexisting continental *crust* and *subcrustal lithosphere*. The latter contribution is comprised of two parts, (a) horizontal recycling that removes a proportion of the total *crust* and *subcrustal lithosphere*, and (b) vertical recycling that removes only a proportion of the *upper crust*. Horizontal recycling represents the areal fraction of *total crust* and *subcrustal lithosphere* consumed by overlapping of tectonic terranes and continental foundering; vertical recycling represents erosional processes, which reduce prior *upper crust* exponentially down to some base level. This erosional base level was taken to be the base of the *upper crust* in earlier versions of the model, but can now be fixed arbitrarily to preserve a veneer

of granitic rock, such as is often found remaining above even old granulitic terranes. For version IV, however, we continue to use the base of the *upper crust* as the base level for vertical recycling.

The *orogene* then undergoes a sequence of internal redistributions of matter, after which it is dispersed to form new *crust* and *subcrustal lithosphere*. The remaining contents of the *orogene* are returned to the *mantle*. In this way new segments of *upper crust*, *lower crust*, and *subcrustal lithosphere* are formed by each orogeny, and are reduced in mass by each subsequent orogeny. This set of operations completes one orogenic cycle. During the interval of time between cycles, isotopic heterogeneity between the major reservoirs is produced by radioactive decay.

Figure 1 is a schematic diagram illustrating the configuration of the model reservoirs and the flow of material (a) toward, (b) within, and (c) away from the *orogene* during the final orogeny. The numbered arrows indicate the sequential algebraic operations, referred to as 'gates', that control the flow of material. Each gate has assigned mass-

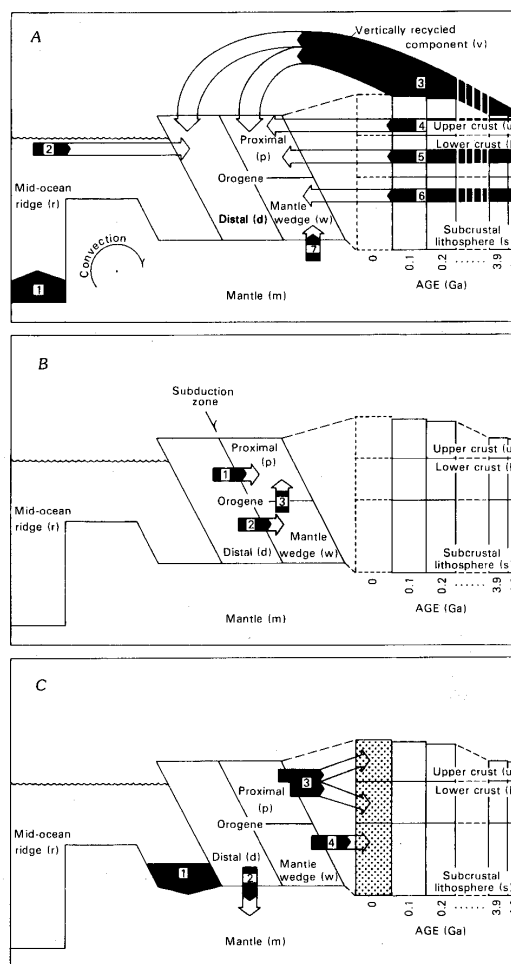


FIG. 1. Final configuration of model reservoirs. The arrows (gates) represent the flow of material (A) toward, (B) within, and (C) away from the *orogene*, which, in turn, produces the stippled block of new *crust* and *subcrustal lithosphere*. Specific function of numbered gates are summarized in Table A1.

and isotope-transport equations and, in the two cases of bifurcating gates, a branching function. It remains only to select the model parameters so that the mass and isotopic configuration of the reservoirs reproduce equivalent terrestrial values.

Consideration of our objectives revealed two shortcomings in earlier versions of plumbotectonics. The large time increments of 400 million years (m.y.) between orogenies did not allow for accurate reproduction of the global orogenic record, and a simple, homogenizing *orogene* did not preserve the internal heterogeneity of an active continental margin. Consequently, the important function of an *orogene* to act not only as a mixer but also as a discriminator, both in time and of matter, was inadequately portrayed. Neither of these deficiencies seriously affect the first-order behavior of the isotopic evolution curves. They do, however, obscure the strikingly episodic nature of orogenies as recorded in the age distribution of rock mass, and they mask chemical biases that arise in the *orogene* itself to cause a decoupling of chemical and isotopic systematics.

We incorporated a finer time-structure into version IV by specifying an *orogene* every 100 m.y. instead of every 400 m.y., as in previous versions. By employing 100 m.y. intervals, the magnitude of each *orogene* can be varied to more realistically mimic the Earth's geologic history (Fig. 2). During periods of high orogenic activity (e.g., Hercynian-Caledonian, Hudsonian, Kenoran, etc.) the size of the *orogene* can be increased by increasing the contribution of *mantle* and *crust* to it. Likewise, at those times, more new *crust* (and *subcrustal lithosphere*) is created relative to quiescent periods. Our reasoning is that during periods of high orogenic activity, more *mantle* is processed at the mid-ocean ridges, supplying, in turn, more oceanic lithosphere to convergent plate boundaries. With more subduction comes greater instability of continents, and greater contribution of recycled *crust* (and *subcrustal lithosphere*) to the *orogene*. For calculation purposes, we maintain a proportionality among *mantle* contribution, horizontal recycling, and mass of the new *crust*. Vertical recycling, however, always removes material from each pre-existing *upper crust* segment at a rate that is related to its height.

Another modification to the model is the subdivision of the *orogene* into three components with specifiable interaction among them. In previous models the *orogene* was represented by a single entity, which completely homogenized all material contributed to it, and which relied on partitioning ratios to redistribute its contents into newly created *crust* (and *subcrustal lithosphere*) and returned *mantle*. Although reasonably successful in producing the desired element concentrations, the totally homogenized *orogene* had lost the ability to preserve any isotopic distinction upon its redistribution. In version

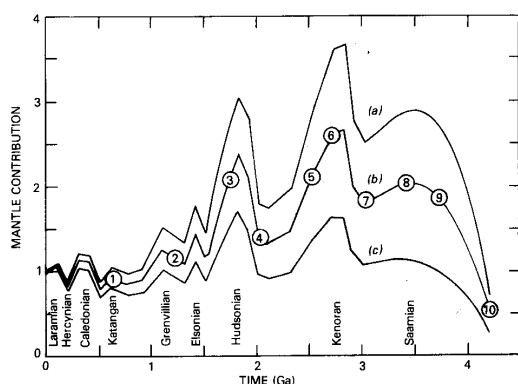


FIG. 2. Mantle contribution to the *orogene* as a function of time normalized to its present-day value. Periods of high orogenic activity as reflected in worldwide distribution of rocks are indicated. Version IV assumes curve (b), while curve (a) is skewed toward early geologic times and curve (c) is skewed toward the present relative to curve (b). Circled numbers on curve (b) indicate, from the present backwards, the number of *mantle* volumes that have been recycled.

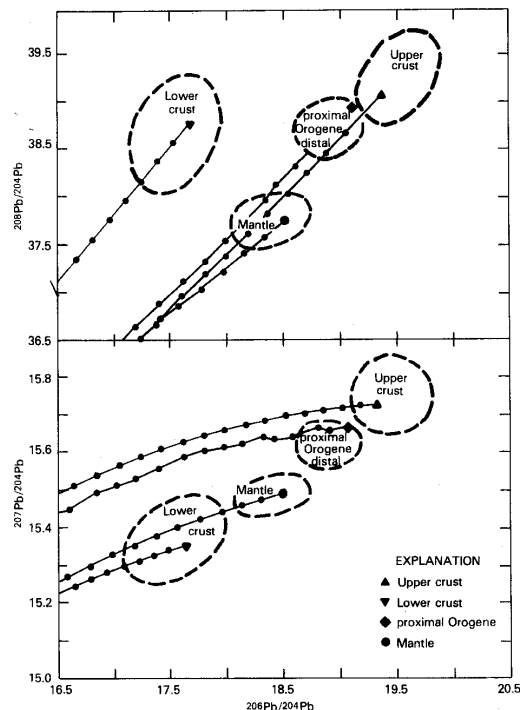


FIG. 3. Lead isotope evolution curves (version IV) for designated model reservoirs together with the fields containing their probable average modern composition (modified from ZARTMAN and DOE, 1981). Dots along each curve indicate progressively older time in 0.1-Ga increments.

IV, we now have an *orogene* that can keep some separate identity among its internal components—a situation deemed necessary to reproduce the chemical and isotopic discriminating nature of its natural counterpart.

Receiving the input of basaltic oceanic crust and subjacent oceanic lithosphere produced from the *mantle* at mid-ocean ridges (gates A1 and A2; Fig. 1) is the *distal* component of the *orogene*, which is located outboard of the subduction zone. A portion of the vertically recycled, or eroded, *upper crust* (gate 3) also is carried as dissolved and suspended matter to the *distal* component of the *orogene*. The remaining portion of the vertically recycled *upper crust* (gate 3) and all of the horizontally recycled *upper crust* (gate 4) and *lower crust* (gate A5) go into the *proximal* component of the *orogene*, where the bulk of sedimentary rocks accumulates inboard of the subduction zone. For our purposes, oceanic sediments that are obducted as accretionary wedges of island arcs without participating in subduction-related volcanism are treated as going directly into the *proximal* component of the *orogene*. Finally, capturing the preexisting *subcrustal lithosphere* that underlies the recycled *crust* (gate A6) and enough additional *mantle* (gate A7) to undergird the newly forming *crust* is the *mantle wedge* component of the *orogene*.

Within the *orogene* partial melting of the subducting *distal* component transports material to the *proximal* (gate B1) and the *mantle wedge* (gate B2) components. The possibility of communication between the *mantle wedge* and *proximal* components (gate B3) is also allowed by the mathematics of the model, but arbitrarily has not been considered. Following this internal redistribution of material, the masses of the *proximal* and *mantle wedge* components exactly equal those of the new *crust* and *subcrustal lithosphere* to be formed. As we have constructed the composite *orogene*, correlation exists

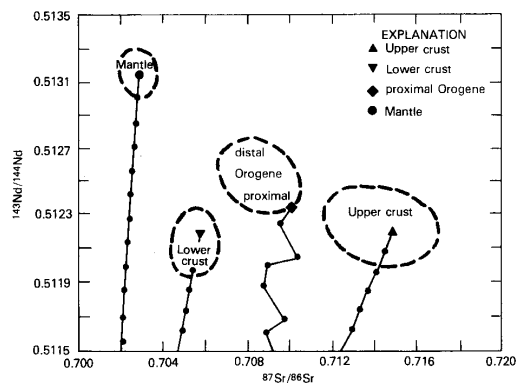


FIG. 4. Strontium-neodymium isotope evolution curves (version IV) for designated model reservoirs together with the fields containing their probable average modern composition (modified from ZARTMAN, 1984a). Dots along each curve indicate progressively older time in 0.1-Ga increments.

between the size of an orogeny and the proportion of *distal* component added to the *proximal* component. This mathematical contrivance creates an isotopic fine-structure in the evolution curves for the *orogene*, reflecting the relative proportions of contributed *crustal* and *mantle* material (Figs. 3, 4). At present, however, we attach little significance to this fine-structure of a model, which certainly oversimplifies geologic processes.

At the conclusion of an orogenic cycle, the part of the *mantle* that was temporarily involved at mid-oceanic ridges in the production of oceanic lithosphere is mixed back into the *mantle* (gate C1). The *orogene* itself is then dispersed with the *distal* component emptying back into the *mantle* (gate C2), the *proximal* component distributing its contents to form the new *upper crust* and *lower crust* (gate C3), and the *mantle wedge* component becoming the new underlying *subcrustal lithosphere* (gate C4). The element partitioning property of gate C3 continues to play a major role in effecting a fractionation of uranium, thorium and lead within the two *crustal* reservoirs. The resultant dichotomous relationship in lead isotopes between upper and lower continental crust remains a keystone to the model.

Ending isotopic compositions for version IV are given in Table 1, and element abundances and concentrations prevailing after the final ($t = 0$) orogeny are given in Table 2. The isotope evolution curves for Pb and for Sr and Nd are plotted in Figs. 3 and 4, respectively, together with the fields containing the probable average composition for modern *mantle*, *upper crust*, *lower crust*, and *orogene* (formerly island arc), as compiled by ZARTMAN and DOE (1981) and ZARTMAN (1984a). In order to portray the isotopic composition of each new *crustal* segment rather than some average of the entire *orogene*, in Figs. 3 and 4 we have replaced the *orogene* curve of previous versions of the model with the curve for only the *proximal* component of the *orogene*. In comparison with earlier modelling efforts, a general similarity, but with some conspicuous exceptions, is noted in the values of the model parameters. The emphasis of the remainder of the paper will be placed upon justifying these several important revisions. The radiogenic isotope evolution curves, for which the model perhaps finds greatest use, are only slightly modified. (See HAINES and ZARTMAN, 1988, for new values of the isotopic ratios of major reservoirs.) That these curves be constrained to fit the observational data is, of course, a fundamental prerequisite of plumbotectonics.

3. Th/U IN THE DEPLETED MANTLE

Strong evidence that the depleted mantle is the recipient of material from another reservoir lies in a long-recognized discrepancy between chemistry and lead isotopes for mid-ocean ridge basalts (MORB). The Th/U of ~ 2.5 as determined chemically for many MORB (SUN, 1980; COHEN and O'NIONS, 1982a; JOCHUM *et al.*, 1983) and a similar value inferred from measured $^{232}\text{Th}/^{238}\text{U}$ for MORB sources (NEWMAN *et al.*, 1983; CONDOMINES *et al.*, 1981; TITAEVA, 1984) contrasts with the Th/U of ~ 3.7 as calculated from measured $^{208}\text{Pb}/^{204}\text{Pb}$ and $^{206}\text{Pb}/^{204}\text{Pb}$ for time-integrated MORB sources (SUN, 1980; COHEN and O'NIONS, 1982a; HAMELIN *et al.*, 1984). Seemingly, the *mantle* can attain its average modern $^{232}\text{Th}/^{238}\text{U}$ value of 2.7 (Table 1) only if some additional source of lead affects its $^{208}\text{Pb}/^{204}\text{Pb}$ - $^{206}\text{Pb}/^{204}\text{Pb}$ systematics.

It appears unlikely that the decoupling of Th/U from the isotopic ratios can be reconciled solely in terms of a mantle fractionation caused by basalt extraction—with the source

Table 1. Ending isotopic composition of reservoirs.

Isotope	Orogene				Upper Crust	Lower Crust	Subcrustal Lithosphere
	Mantle	Distal	Proximal	Mantle Wedge			
$^{206}\text{Pb}/^{204}\text{Pb}$	18.47	18.62	19.06	18.40	19.33	17.62	18.32
$^{207}\text{Pb}/^{204}\text{Pb}$	15.48	15.52	15.67	15.46	15.73	15.35	15.44
$^{208}\text{Pb}/^{204}\text{Pb}$	37.73	37.96	38.94	38.04	39.07	38.75	38.10
$^{238}\text{U}/^{238}\text{U}$	10.01	10.56	10.28	9.88	11.08	6.49	9.13
$^{232}\text{Th}/^{238}\text{U}$	2.73	2.59	4.17	3.36	3.88	6.05	3.89
$^{87}\text{Sr}/^{86}\text{Sr}$	.7029	.7031	.7102	.7037	.7149	.7057	.7042
$^{87}\text{Rb}/^{86}\text{Sr}$	.040	.051	.279	.071	.456	.048	.077
$^{143}\text{Nd}/^{144}\text{Nd}$	.51314	.51313	.51232	.51290	.51219	.51219	.51272
^6Nd	+9.8	+9.6	-6.2	+5.2	-8.7	-8.7	+1.6
$^{147}\text{Sm}/^{144}\text{Nd}$	.222	.208	.167	.193	.165	.170	.193

Table 2. Ending element abundances and concentrations.

		Mantle	Upper Crust	Lower Crust	Subcrustal Lithosphere
Mass ($\times 10^{24}$ g)		1000.1	6.70	15.91	27.28
Abundances ($\times 10^{15}$ moles)	^{204}Pb	2.00	9.95	6.66	0.386
	^{238}U	20.1	110	43.3	3.53
	^{232}Th	54.7	428	262	13.7
	^{86}Sr	11700	2850	6960	461
	^{87}Rb	469	1300	337	35.5
	^{144}Nd	1340	356	356	48.0
Average Concentration (ppm)	^{147}Sm	297	58.6	60.6	9.29
	Pb	.030	23.1	6.30	.214
	U	.0048	3.94	.653	.0310
	Th	.0127	14.8	3.82	.117
	Sr	10.40	378	389	15.0
	Rb	.144	59.5	6.50	.399
	Nd	.810	32.1	13.5	1.06
	Sm	.296	8.72	3.80	.340

Th/U changing from 5–6 to ~ 2.5 over the lifetime of the Earth. Such a lowering of Th/U in the mantle should be accompanied by an equivalent trend in the crust. TAYLOR and MCLENNAN (1985) show that Th/U in sedimentary rocks probably has increased with time, and if reflecting the behavior of the crust in general, this trend is opposite to that predicted for a simple fractionating mantle. Recently, GALER and O'NIONS (1985) have sought to explain the discrepancy in terms of different residence times for U, Th, and Pb in the upper mantle as these elements passed from a source in the lower mantle to the crust. Because of the required short upper mantle residence times, the disparity in $^{207}\text{Pb}/^{204}\text{Pb}$ observed between the crust and upper mantle was thought to rule out crustal recycling as an important mechanism.

We propose that preferential recycling of uranium relative to thorium from the crust back into the mantle does resolve the problem and is compatible with all observational data. The survival of old *upper crustal* segments and a chemically discriminating *orogene* can effect the desired decoupling over a time scale sufficiently long to maintain the $^{207}\text{Pb}/^{204}\text{Pb}$ distinction between crust and mantle. Accordingly, we are able to explain the enigma without appealing to the lower mantle.

The needed element bias has been known almost since the advent of modern marine geochemistry. The extreme insolubility of thorium virtually excludes it from seawater (SACKETT *et al.*, 1958; LI, 1982; CHEN *et al.*, 1986a), and this absence of a dissolved component is manifest in the thorium content of bulk deep-sea sediments, which are comprised of biochemically-precipitated oozes (Th/U ≈ 0) and pelagic clays (Th/U ≈ 5). On the other hand, uranium is largely oxidized to U^{+6} in the weathering cycle, so that its concentration in seawater as a carbonate complex is several orders of magnitude higher than that of thorium. The ability of uranium to be carried by convecting hydrothermal systems into underlying basalts of the oceanic crust further diminishes Th/

U in those rocks most likely to be subducted as eclogites back into the mantle (MICHARD and ALBARÈDE, 1985; CHEN *et al.*, 1986b). The overlying sediments appear to be largely scraped off or melted at the convergent plate margin and incorporated into the newly-forming island arc instead of being returned to the mantle.

Our model takes advantage of this geochemical bias by discriminating against the return flow of Th from the *erosional* component of the *upper crust* to the *distal* component of the *orogene* (gate A3; Fig. 1)—reflecting the arrest of both dissolved and heavy-mineral hosted Th at the continental edge. In contrast, the corresponding return flow of U is favored—reflecting this element's solubility in seawater and enhanced retention in the hydrothermally altered basalt. The return flow of Pb—containing thorogenic ^{208}Pb produced while resident in the crust—is presumed to occur without discrimination, and imparts on the *mantle* an isotopic composition only slightly different from that of "bulk Earth". Fractionation factors for Pb, U, and Th have been arbitrarily assigned values of 1, 1.5, and 0.67, respectively (Table A2), although other combinations were found to work equally well. The values for the fractionation factors that we have chosen would prevail if roughly half of the returned crustal U derived from seawater and the other half from pelagic sediments. A closer consideration of Sr and Nd isotopic systems may suggest a discrimination in their return flow, too, but such a treatment is beyond the scope of this paper.

The effectiveness of the *distal* component of the *orogene* in driving down the Th/U in the *mantle* is shown in Fig. 5. (Figure 5 also shows the corresponding behavior of the U/Pb in the *mantle*, which, after initially decreasing, increases in value due to the preferential return of uranium relative to lead. Although the anomalously high $^{238}\text{U}/^{204}\text{Pb}$ of 10.0 in the present-day *mantle* can be avoided by reducing the U/Pb bias, the wisdom of basing the mantle $^{238}\text{U}/^{204}\text{Pb}$ on closed-

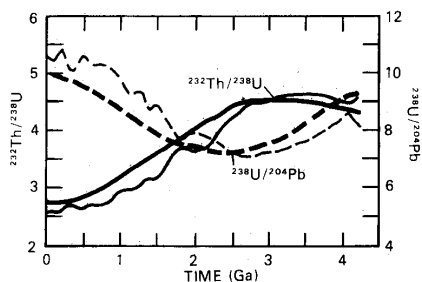


FIG. 5. $^{232}\text{Th}/^{238}\text{U}$ and $^{238}\text{U}/^{204}\text{Pb}$ in the mantle (heavy lines) and distal component of the orogene (light lines) as a function of time (version IV).

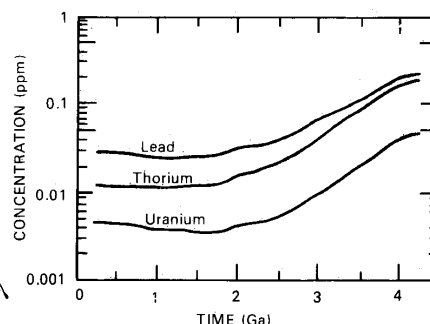


FIG. 7. Concentration of U, Th, and Pb in the mantle as a function of time (version IV).

system calculations is certainly brought into question.) At the same time, the necessary distinction in $^{207}\text{Pb}/^{204}\text{Pb}$ is maintained (Table 1 and Fig. 3) by appropriate selection of parameters governing the transport of lead from the upper crust to the distal component of the orogene. Reasonable values of the mass fraction of recycled upper crust, $\Phi^{\Sigma v \rightarrow d,p}$, and the proportion of that mass going into the distal component of the orogene, $\Delta^{d,p}$, are shown as a stippled area in Fig. 6. Not surprisingly, permitted combinations of $\Phi^{\Sigma v \rightarrow d,p}$ and $\Delta^{d,p}$ produce an essentially constant mass flux from the upper crust to the distal component of the orogene.

In order to effect the desired decoupling of chemical and isotopic ratios, the residence times must be short enough and, thus, the concentrations of Pb, U, and Th in the mantle must be low enough to respond to the component of recycled material. Adjusting the concentration levels of these elements in the mantle (Table 2 and Fig. 7) to meet this new constraint represents the most significant change over earlier versions. Although we differ with some aspects of GALER and O'NIONS (1985), we do accept their estimate of 20–30 ppb for the present-day concentration of lead in the depleted mantle. ZINDLER and HART (1986) have recently given values for the concentrations of Pb and U in the depleted mantle that

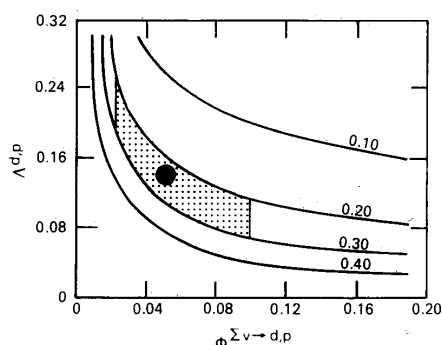


FIG. 6. $\Phi^{\Sigma v \rightarrow d,p}$ versus $\Delta^{d,p}$ showing difference in ending $^{207}\text{Pb}/^{204}\text{Pb}$ for upper crust and mantle reservoirs. Stippled area is considered to represent geologically and isotopically reasonable values of the parameters (values used for version IV indicated by ●).

are about two times higher, but this general convergence of estimates within the range of 20 to 60 ppb is almost certainly more realistic than the value of 340 ppb given in the previous plumbotectonics model (ZARTMAN, 1984a).

4. GEOCHRON AND THE AGE OF THE EARTH

A powerful constraint imposed by the coupled decays of ^{238}U ($\rightarrow^{206}\text{Pb}$) and ^{235}U ($\rightarrow^{207}\text{Pb}$) requires that the total Earth's lead isotopic composition must lie on the geochron; that is, on an isochron passing through the primordial lead isotopic composition and having a slope equivalent to the age of the Earth. Knowledge of the present-day $^{206}\text{Pb}/^{204}\text{Pb}$ and $^{207}\text{Pb}/^{204}\text{Pb}$ for the total Earth allows the calculation of the age of the Earth, assuming that we also know the primordial values of these ratios. Numerous workers have observed that most of the studied terrestrial reservoirs, such as the depleted mantle and, especially, the upper continental crust, have isotopic compositions that lie to the right of the geochron on a $^{207}\text{Pb}/^{204}\text{Pb}$ - $^{206}\text{Pb}/^{204}\text{Pb}$ diagram as calculated by using Canon Diablo meteorite as primordial lead and an Earth age of 4.55 Ga. This precise age has been obtained from the dating of chondritic meteorites, and is, of course, only indirectly extrapolated to the Earth. That the depleted mantle should show an apparent increase in U/Pb with time is contrary to a more incompatible nature of uranium than lead, and has received considerable attention as the so-called lead paradox (ZINDLER and HART, 1986, and references therein).

Possible locations of the missing reservoir(s) with complementary lowered U/Pb have been proposed to be the Earth's core (VOLLMER, 1977; VIDAL and DOSSO, 1978; ALLÈGRE *et al.*, 1982) and the lower continental crust (O'NIONS *et al.*, 1979; ZARTMAN, 1984b). NEWSOM *et al.* (1986) have argued against a gradual growth of the Earth's core based on the behavior of siderophile and chalcophile elements in the mantle, and, instead, present a lead isotopic mass balance involving only the crust and mantle, which plots on the 4.55 Ga geochron. We believe, however, that they assumed too low values of $^{206}\text{Pb}/^{204}\text{Pb}$ for both the upper and lower crust in making their calculation. Even the isotopic composition for crust and mantle proposed by ZARTMAN (1984b) to re-

solve the paradox for a 4.50 Ga Earth now represents an uncomfortable fit to the observed data.

The possibility that the Earth is significantly younger than chondritic meteorites does offer another plausible explanation of the discrepancy. Calculations of the accretion time of a planetary body permit an interval of several tens of millions of years during which collision heating might cause volatile element loss (SAFRONOV, 1969; WETHERILL, 1980). An increase in U/Pb from its chondritic value of <0.01 to its terrestrial value of ~ 0.2 occurring 50 m.y. after chondrite condensation would have a noticeable effect on lead isotopic evolution. A misfit of about a hundred million years to the 4.55 Ga geochron likely exists, and some additional explanation seems to be required (ALBARÈDE and JUTEAU, 1984).

The sensitivity of the age of the Earth, T_0 , to the total Earth lead isotopic mass balance is apparent in Fig. 8, where the positions of average isotopic compositions for the major reservoirs are shown relative to a family of geochrons spanning the age range from $T_0 = 4.60$ Ga to 4.30 Ga. Although recognition of the lower continental crust with its retarded lead isotopic evolution favors a somewhat older age of the Earth than would be deduced otherwise, the most likely age to be calculated from our model is 4.45 Ga. The maximum age that appears to be compatible with any closed system model is about 4.50 Ga. Thus, if an age of the Earth approaching that of the 4.55 Ga age of chondritic meteorites is to be entertained, an additional reservoir, such as the lower mantle or core, or some mechanism such as lead loss during planetary accretion or, possibly, the fission of the Moon (TAYLOR, 1987), is required. Note, however, that this isolation may not occur 'instantaneously' at 4.55 Ga before any terrestrial lead isotopic evolution has taken place. We make no attempt to identify specifically and account for the missing lead in our modelling effort; instead, we begin our model arbitrarily at 4.45 Ga with an initial isotopic composition equal to Canon Diablo lead. Subsequently, all matter resides in an undifferentiated mantle until 4.2 Ga, after which crustal separation commences and isotopic heterogeneity among reservoirs is established.

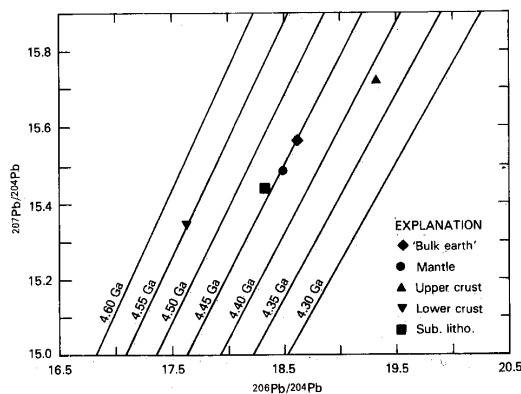


FIG. 8. Average lead isotopic composition for designated model reservoirs relative to geochrons for $T_0 = 4.60, 4.55, 4.50, 4.45, 4.40, 4.35$, and 4.30 Ga. Initial lead taken to be that of Canon Diablo meteorite.

This uncertainty surrounding the origin of the Earth may affect the isotopic evolution of strontium as well. In particular, if significant time did elapse during planetary accretion, accompanied by a gradual decrease in Rb/Sr from its chondritic value of ~ 0.3 to its terrestrial value of ~ 0.03 , we must entertain the possibility of a considerably more isotopically evolved strontium at the onset of mantle differentiation. The same 50 m.y. delay in acquiring a terrestrial U/Pb mentioned above could result in a bulk Earth $^{87}\text{Sr}/^{86}\text{Sr}$ at 4.2 Ga of 0.7000 instead of 0.6994, which one calculates without such delay. Although we do not use a more evolved strontium in version IV, further attention needs to be given to this interesting problem. The refractory nature of the rare earth elements makes a corresponding large change in Sm/Nd due to volatile element loss very unlikely. In contrast to volatilization processes, the scavenging effect of core separation on lead would not be expected similarly to disturb strontium or neodymium isotopic systematics. Although leaving unresolved important aspects of earliest Earth history, these possible discrepancies in the model do not seriously affect other findings of the paper.

5. CRUSTAL RECYCLING

A question under debate is just how much crustal recycling back into the mantle occurs in island-arc systems on a global scale (TAYLOR and MCLENNAN, 1985; ALLÈGRE and ROUSSEAU, 1984; WHITE and PATCHETT, 1984; COHEN and O'NIONS, 1982b; ARMSTRONG, 1981; MOORBATH, 1978)? Geochemical and isotopic studies of various subduction regimes have reached different conclusions concerning the importance of sediments either as a component of arc-derived magmas or as a longer-term contaminant of the mantle. Even where evidence for the subduction of sedimentary rock is minimal, opportunity may exist for oceanic crust to acquire elements ultimately derived from the continents. Hydrothermal alteration causing the uptake of uranium by submarine basalt has already been suggested as a mechanism to create a Th/U bias between crust and mantle.

We must thus ask if permissible rates of recycling can produce the crustal signature in the mantle that is required by the plumbotectonic model. To lend physical reality to version IV, we can translate its numerical parameters into the following plate tectonic scenario. During the most recent ($t = 0$) orogeny, about one-sixth of the upper mantle is brought into the mid-ocean ridge environment. Here is produced an oceanic lithosphere 25 km thick, equal in area to that of the ocean basins, and spanning 100 m.y. in age. The incompatible element concentration assumed for this lithosphere is equivalent to a basal 20 km of highly depleted harzburgite overlain by 5 km of basalt and gabbro. Excluding material to be directly returned to accretionary wedges, a 30 m veneer of deep-sea sediments, which represents 14 percent of the eroded sedimentary load from the continents, creates the top layer of our 'geochemical' oceanic lithosphere. Next, hydrothermal alteration selectively introduces dissolved incompatible elements into the basalt as it traverses the ocean basin. Although some of the incompatible elements—approximately $\frac{1}{4}$ in version IV—will be transferred to the continent by subduction-related magmatism, a return mass flux of approximately

3×10^{17} g/yr (10^2 km³/yr) to the mantle carries with it an isotopic and geochemical signature of the entrained sediments and/or altered basalt. We see nothing in this scenario that is inconsistent with the kinetics of plate tectonics.

Paramount to our model as now construed is the discrepancy between Th/U of the depleted mantle as inferred from MORB chemistry and as calculated from its $^{208}\text{Pb}/^{204}\text{Pb}$ and $^{206}\text{Pb}/^{204}\text{Pb}$. Intuitively, one can recognize that such decoupling requires an integrated flux of these elements back into the mantle commensurate with their abundances in the mantle. Furthermore, a sufficient concentration gradient must exist between the crust and mantle to overcome the preponderance in mass of the mantle. That the concentration of lead in the mantle is probably an order of magnitude lower than the estimate of ZARTMAN (1984a)—in the range of 20 to 60 ppb—has already been discussed. The plausibility of plumbotectonics depends heavily on taking advantage of this concentration leverage, which places approximately 90 percent of the lead in the crust. Less depletion of the mantle will rapidly produce a convergence of the chemically and isotopically determined values of Th/U.

Several variants of the model can be explored in order to assess the role of crustal recycling, and to place limits on its extent. Outside permissible bounds, the model either will fail to yield appropriate isotopic systematics or will violate some other geochemical or geologic constraint. One extreme would be to minimize the amount of crustal material allowed to be subducted back into the mantle. Obviously, to shut this process down completely would be unacceptable, at least to the survival of bi-directional transport. However, we may restrict ourselves to no loss of continental area, and allow only the vertical recycling function to operate. Accordingly, mantle differentiation builds the crust with an areal distribution of rock ages equal to their final, present distribution. Because the vertical recycling function extracts material only from the upper crust, which already has a lower Th/U than the total crust, there may be some advantage in favoring such restricted recycling. This variant delays crust formation as long as possible, and a contraction between the isotopic fields of reservoirs might be expected. Apparently, the accompanying reduction in the return flux to the mantle partially counteracts this effect.

An opposite extreme would be a variant of the model having maximum crustal recycling. A geologically reasonable upper limit to recycling is uncertain, but we have examined a number of situations ranging from that of no areal reduction to that of 50 percent areal reduction; that is, over the lifetime of the Earth twice the present-day continental area is created. With minor adjustments of element distribution coefficients, all of these model variants could be brought into reasonable agreement with the observed data. It is something of a compromise, therefore, that we choose for version IV a 33 percent areal reduction; that is, over the lifetime of the Earth 1.5 times the present-day continental area is created (Fig. 9).

Another means to explore the timing of crustal recycling is by keeping the percent of areal reduction constant while varying the relative magnitude of orogenies. No particular advantage was seen in changing the details of the distribution of orogenies, but skewing the overall pattern to emphasize either young or old orogenies merits investigation, particularly

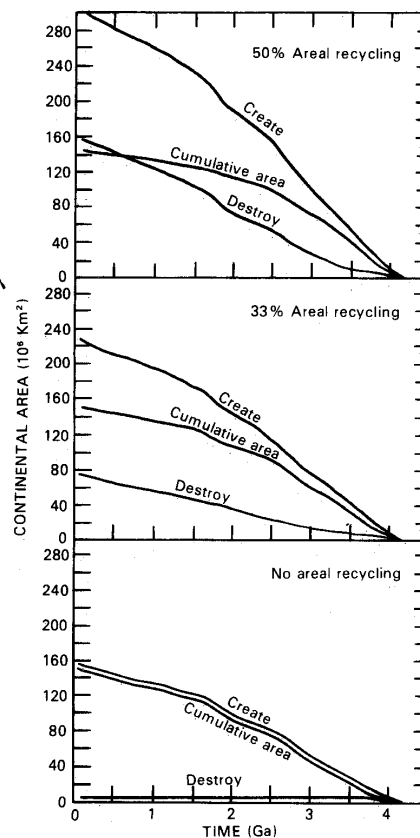


FIG. 9. 50%, 33%, and 0% areal recycling of the crust to produce a final continental area of approximately 150×10^6 km². Shown is the total area created, the area destroyed by recycling, and the cumulative remaining area.

in light of our general ignorance about the distant geologic past. A skewing function was applied, which allowed orogenic activity to range from a comparable magnitude over all of geologic time to an order of magnitude enhancement for earliest geologic time. Again, we find an insensitivity of the model parameters to the relative magnitude of the orogenies, although the $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ slope based on the isotopic ratios of the modern reservoirs reflect weakly the changing mean age of activity. In skewing orogenic activity toward the present, we ultimately would face the problem of producing too high a $^{87}\text{Sr}/^{86}\text{Sr}$ in the mantle because of the delayed transfer of rubidium to the crust. Version IV here, too, adopts a compromise stand on the relative magnitude of orogenies as a function of time (Fig. 2).

We proceed to address an aspect of crustal differentiation that does have a profound effect on the model. The element distribution coefficients govern the rate of transfer of each species between reservoirs. As such, the distribution coefficients strongly control the relative abundances of the elements in the different reservoirs, and determine the concentrations

toward which the elements are driven upon approach to a steady-state crustal configuration. This consideration, not fully exploited in previous versions, now takes on major significance as we attempt a decoupling of chemistry and radiogenic isotopes. No other model parameter can reasonably be used to achieve the desired disparity between Th/U and $^{208}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$.

The consequence of varying the element distribution coefficients in order to effect a factor of two increase or decrease in the concentration of lead in the *mantle* is portrayed in Fig. 10. With a modern *mantle* lead content of 60 ppb we lose over half of the desired disparity between actual and apparent Th/U; a lead content of 15 ppb produces only a small increase in disparity. The decoupling is time-dependent, and Th/U eventually approaches a steady-state value determined by the element partitioning between the *distal* and *proximal* components of the *orogene*. A stronger partitioning of Th and U or a lower lead content of the *mantle* is not supported by available data, and, accordingly, we are constrained in the selection of element distribution coefficients. Although the model allows considerable flexibility in configuring the mass of crustal growth and recycling, it places tight limits on the relative concentration levels in the reservoirs. Based on the better-known crustal concentrations, we obtain rather narrow limits of 20 to 40 ppb lead in the *mantle*. Whether our model *mantle* adequately describes an equivalent reservoir in the Earth will be further considered in the following section.

As seen in Fig. 10, an important clue concerning the separation of the *crust* from the *mantle*, and, by analogy, the past pattern of orogenic activity, lies in the behavior of the Th/U in the *mantle* as a function of time. Increasing the extent or advancing the timing of crustal differentiation will accelerate the decoupling of the Th/U and bring down its *mantle* value earlier. Interestingly, version IV shows most of the decrease in Th/U to take place since the Late Archean. Because the Th/U decoupling only begins after *mantle* depletion is essentially complete, a significantly earlier decrease in the ratio will require a substantially earlier crustal differentiation. If reliable Th/U measurements can be obtained for the Proterozoic and Archean *mantle*, we may be able to further constrain our model by tracing this parameter through geologic time. Obtaining time dependent data for thorium and uranium within all major reservoirs could well be a very fruitful future pursuit.

6. MANTLE HETEROGENEITY

The intrinsic isotopic heterogeneity of the Earth's *mantle*, both as reflected within the population of MORB and among oceanic-island basalts, is not being denied nor ignored. Rather, we believe that for our purposes the *mantle* hetero-

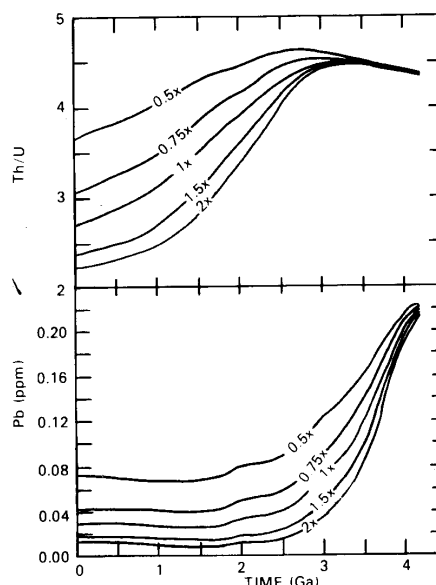


FIG. 10. Th/U and Pb concentration in the *mantle* as a function of time for element distribution coefficients 0.5 $\times$, 0.75 $\times$, 1 $\times$, 1.5 $\times$, and 2 $\times$ those of version IV.

geneity can be represented by an average isotopic composition and element concentration. Although our model *mantle* assumes the dominance of a depleted MORB source, we also acknowledge the presence of other components in making these average assignments.¹ The simplistic treatment of the *mantle* as a single, homogeneous reservoir requires the validity of two assumptions.

Firstly, the acknowledged heterogeneity of the *mantle* either (1) arises from a minor amount of oceanic-island basalt sources relative to a preponderant, depleted MORB source, and/or (2) is short-lived relative to the age of the Earth. Taking the $^{207}\text{Pb}/^{204}\text{Pb}$ vs. $^{206}\text{Pb}/^{204}\text{Pb}$ slope of 0.10 (equivalent to a mean age of about 1.6 Ga) displayed by mid-ocean ridge and oceanic-island basalts worldwide as evidence of isolation times appreciably shorter than the age of the Earth, we are probably justified in this assumption even if the oceanic-island basalt sources are not insignificant. A preliminary attempt to introduce some *mantle* heterogeneity into the model by delaying for a comparable time the mixing of returned material from the *orogene* was found to have little impact on the general behavior of the model. Furthermore, during each orogenic cycle the *orogene* receives a volume of oceanic lithosphere commensurate with that of all the ocean basins, so, to the extent that this lithosphere is isotopically representative of the upper *mantle*, we have the desired average sample required by the model.

Secondly, subsequent to some very early differentiation of the total planet, no additional hidden or unaccounted-for reservoir plays a significant role in determining the chemical or isotopic makeup of the upper *mantle* and *crust*. That is, if transport to or from the lower *mantle* or core continues to

¹ For example, a *mantle* comprised of (1) 70 percent N-type MORB source (Pb = 20 ppb, U = 3 ppb, and Th = 7 ppb, and $^{206}\text{Pb}/^{204}\text{Pb}$ = 18.1, $^{207}\text{Pb}/^{204}\text{Pb}$ = 15.45, and $^{208}\text{Pb}/^{204}\text{Pb}$ = 37.5), (2) 20 percent P-type MORB source (Pb = 35 ppb, U = 5 ppb, and Th = 15 ppb, and $^{206}\text{Pb}/^{204}\text{Pb}$ = 18.4, $^{207}\text{Pb}/^{204}\text{Pb}$ = 15.48, and $^{208}\text{Pb}/^{204}\text{Pb}$ = 37.8), and (3) 10 percent ocean island basalt sources (Pb = 100 ppb, U = 20 ppb, and Th = 70 ppb, and $^{206}\text{Pb}/^{204}\text{Pb}$ = 19.0, $^{207}\text{Pb}/^{204}\text{Pb}$ = 15.54, and $^{208}\text{Pb}/^{204}\text{Pb}$ = 38.2) would have the required average composition predicted by our model.

effect the isotopic budget, modification or even abandonment of plumbotectonics as we have construed it may be necessary. Essentially, we take the position that if a chemical and isotopic material balance based on observation of the 'visible' part of the Earth can be made, we should examine such a model before appealing to a less testable scenario.

Geophysical evidence is still not definitive regarding total- or layered-mantle convection, or the final fate of subducted oceanic crust. However, of the competing Earth models, we believe that a strong case can be made for a chemically-stratified mantle, such as proposed by ANDERSON (1987, and references therein). In particular, kinetic considerations cause us to limit the *mantle* to that region above the 670 km seismic discontinuity. The exchange rates and mantle turnover times needed in order to successfully model the observed terrestrial reservoirs are quite reasonable when we consider the upper mantle only, but may pose serious problems with the total mantle. If, as is generally held, the depleted upper mantle is chemically complementary to the continental crust, how can this composition be retained with rapid total mantle convection? How could this depleted signature be transferred to the upper mantle by subducting oceanic crust that returns to the very base of the mantle?

In the preceding paragraph mention was made of *mantle* turnover times. We now look further at this aspect of the model, which, in combination with the element distribution coefficients, affects the timing of crustal differentiation. For our model we must specify the total mass fraction of *mantle* involved in producing oceanic crust (passing through Gate A1; Fig. 1) over the lifetime of the Earth. This mass fraction may not exceed unity for any single orogeny, but the summation may be as much as ~20 times the mass of the *mantle* before that condition is violated. The necessity of driving the *mantle* to a high degree of depletion was already discussed in the section on Crustal Recycling, and it is obvious that a minimum number of turnovers must also occur before depletion approaches steady-state. The effect of choosing mass fractions summing to 4, 6, 8, 10, 12, and 14 recyclings of the *mantle* on Th/U and lead concentration in the *mantle* is illustrated in Fig. 11. The choice for version IV of 10 recyclings of the *mantle* is, again, a compromise between insufficient interaction to satisfy geochemical and isotopic requirements, and implausibly high rates of convective movement. Recycling the total mantle 10 times seems excessive, and becomes one of our arguments against the involvement of the lower mantle except in earliest geologic times.

In line with the assumption that significant mantle heterogeneity does not persist longer than about 1.6 Ga, it is noteworthy that, with a total of 10 recyclings, the *mantle* for version IV would have undergone one recycling since 0.6 Ga, two recyclings since 1.2 Ga, and three recyclings since 1.7 Ga (Fig. 2). Although, admittedly, considerable uncertainty exists about the sources giving rise to oceanic-island basalts, the likelihood of their survival beyond three *mantle* recyclings seems small. Random sampling theory predicts that after 1 recycling, 37 percent, after 2 recyclings, 13 percent, and after 3 recyclings, 5 percent of the population would escape selection. Apparently, much of the observed mantle heterogeneity was produced during and subsequent to the last(?) period of profound orogenic activity and crustal growth having expres-

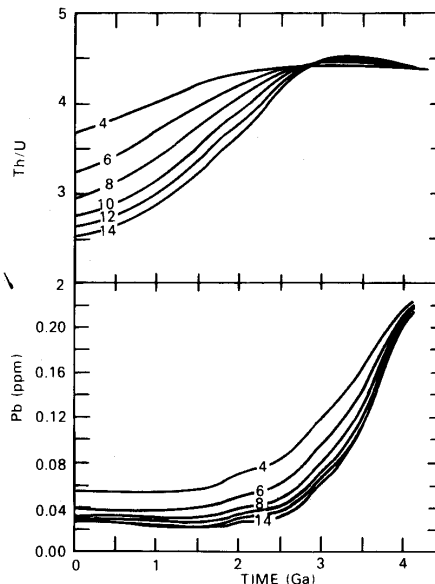


FIG. 11. Th/U and Pb concentration in the *mantle* as a function of time for 4, 6, 8, 10, 12, and 14 recyclings of the *mantle*.

sion on every continent—the 1.6–1.9 Ga Hudsonian Orogeny.

The implied linking between orogenic activity on the continents and the lifetime of mantle heterogeneity certainly deserves further attention. Here we mention only one implication of plumbotectonics to this heterogeneity. The depleted mantle giving rise to MORB must result mainly from the return to the *mantle* of the *distal* component of the *orogene*. As such, our model requires that subducted oceanic lithosphere, after modification on the ocean floor and extraction of material into the island arc, forms the bulk of the mantle lying above the 670 km seismic discontinuity. This complement to the continental *crust* bears the geochemical and isotopic signature of MORB, and is of sufficient volume to produce them. Oceanic-island basalt sources, on the other hand, are more enigmatic. It has been suggested that they originate as a cognate to subducted oceanic crust, from detached fragments of the subcontinental mantle, by magmatic and/or metasomatic processes localized elsewhere in the depleted upper mantle, or by diapiric(?) injections from the lower mantle. The review of the geochemical and isotopic characteristics of mantle-derived volcanic rocks by ZINDLER and HART (1986), for example, attempts to identify such various source end-members. For now, we speculate that the ultimate answers will not be fatal to the plumbotectonic model.

Finally, a brief mention of the *subcrustal lithosphere*, that part of the mantle underlying the continental crust and permanently attached to it. This continental keel is now known to be a reservoir displaying quite different isotopic characteristics than, and having a history unique from, the depleted, asthenospheric *mantle*. The *subcrustal lithosphere* may sometimes be involved in the transfer of matter from the

orogene to newly forming *crust*, but probably doesn't dominate in the process. Old *subcrustal lithosphere* is most obvious because of the distinctive isotopic composition it imparts to basaltic rocks that later penetrate the continental crust. Despite much ignorance about this reservoir, its apparent similar size to the continental crust demands its inclusion in the model. Each new segment of *subcrustal lithosphere* is created in the *mantle wedge* component of the *orogene*, and has been modelled as a mixture of preexisting *subcrustal lithosphere*, captured *mantle*, and 1 percent of introduced matter from the *distal* component of the *orogene* (the former two in proportion to the area of new *crust* that replaces preexisting *crust*, and the area that represents expansion over previous oceanic crust, respectively). In addition to its importance for explaining the isotopic composition of continental volcanic rocks, detached portions of the continental keel may contribute to the heterogeneity of the *mantle*. A return of the *subcrustal lithosphere* to the *mantle*, however, has not been included in the model because of its presumed minor contribution to the isotopic budget.

Acknowledgements—I wish to thank K. R. Ludwig, B. R. Doe, S. B. Jacobsen, W. M. White, and J. L. Rubenstone for reviewing an earlier draft of the manuscript and offering helpful suggestions to improve the plumbotectonic model. Advice to the senior author by G. J. Wasserburg to take a break occasionally from the laboratory and think about the big picture has been a major stimulus to this effort.

Editorial handling: D. A. Papanastassiou

REFERENCES

- ALBARÈDE F. and JUTEAU M. (1984) Unscrambling the lead model ages. *Geochim. Cosmochim. Acta* **48**, 207–212.
- ALLÈGRE C. J. and ROUSSEAU D. (1984) The growth of the continent through geological time studied by Nd isotope analysis of shale. *Earth Planet. Sci. Lett.* **67**, 19–34.
- ALLÈGRE C. J., DUPRÉ B. and BREVART O. (1982) Chemical aspects of the formation of the core. *Phil. Trans. Roy. Soc. London* **306**, 49–59.
- ALLÈGRE C. J., HAMELIN B., PROVOST A. and DUPRÉ B. (1987) Topology in isotopic multispace and origin of mantle chemical heterogeneities. *Earth Planet. Sci. Lett.* **81**, 319–337.
- ANDERSON D. L. (1987) The depths of mantle reservoirs. *Geochem. Soc., Spec. Publ.* **1**, 3–12.
- ARMSTRONG R. L. (1981) Radiogenic isotopes: the case for crustal recycling on a near steady-state no-continental-growth Earth. *Phil. Trans. Roy. Soc. London* **A301**, 443–472.
- CHEN J. H., EDWARDS R. L. and WASSERBURG G. J. (1986a) ^{238}U , ^{234}U , and ^{232}Th in seawater. *Earth Planet. Sci. Lett.* **80**, 241–251.
- CHEN J. H., WASSERBURG G. J., VON DAMM K. L. and EDMOND J. M. (1986b) The U-Th-Pb systematics in hot springs on the East Pacific Rise at 21°N and Guaymas Basin. *Geochim. Cosmochim. Acta* **50**, 2467–2479.
- COHEN R. S. and O'NIONS R. K. (1982a) The lead, neodymium and strontium isotopic structure of ocean ridge basalts. *J. Petrol.* **23**, 299–324.
- COHEN R. A. and O'NIONS R. K. (1982b) Identification of recycled continental material in the mantle from Sr, Nd, and Pb isotopic investigations. *Earth Planet. Sci. Lett.* **61**, 73–84.
- CONDOMINES M., MORAND P. and ALLÈGRE C. J. (1981) ^{230}Th - ^{238}U radioactive disequilibrium in tholeiites from the FAMOUS zone (Mid-Atlantic Ridge, 36°50'N). *Earth Planet. Sci. Lett.* **55**, 247–256.
- DOE B. R. (1967) The bearing of lead isotopes on the source of granitic magma. *J. Petrol.* **8**, 51–83.
- DOE B. R. and ZARTMAN R. E. (1979) Plumbotectonics I, The Phanerozoic. In *Geochemistry of Hydrothermal Ore Deposits*, 2nd edn. (ed. H. L. BARNES), Chap 2, pp. 22–70, Wiley Interscience.
- GALER S. J. G. and O'NIONS R. K. (1985) Residence time of thorium, uranium and lead in the mantle with implications for mantle convection. *Nature* **316**, 778–782.
- HAINES S. M. and ZARTMAN R. E. (1988) A Hewlett-Packard Series 200/300 BASIC language program for version IV of plumbotectonics. *U.S. Geol. Surv. Open-File Rept.* **88-269**, 79p.
- HAMELIN B., DUPRÉ B. and ALLÈGRE C. J. (1984) Lead-strontium isotopic variations along the East Pacific Rise and the Mid-Atlantic Ridge: A comparative study. *Earth Planet. Sci. Lett.* **67**, 340–350.
- JOCHUM K. P., HOFMANN A. W., ITO E., SEUFERT H. M. and WHITE W. M. (1983) K, U and Th in mid-ocean ridge basalt glasses and heat production, K/U and K/Rb in the mantle. *Nature* **306**, 431–436.
- LI Y.-H. (1982) A brief discussion on the mean oceanic residence time of elements. *Geochim. Cosmochim. Acta* **46**, 2671–2675.
- MICHARD A. and ALBARÈDE F. (1985) Hydrothermal uranium uptake at ridge crests. *Nature* **317**, 244–245.
- MOORBATH S. (1978) Age and isotope evidence for the evolution of continental crust. *Phil. Trans. Roy. Soc. London* **A288**, 401–413.
- NEWMAN S., FINKEL R. C. and MACDOUGALL J. D. (1983) ^{230}Th - ^{238}U disequilibrium systematics in oceanic tholeiites from 21°N on the East Pacific Rise. *Earth Planet. Sci. Lett.* **65**, 17–33.
- NEWSOM H. E., WHITE W. M., JOCHUM K. P. and HOFMANN A. W. (1986) Siderophile and chalcophile element abundances in oceanic basalts, Pb isotope evolution and growth of the Earth's core. *Earth Planet. Sci. Lett.* **80**, 299–313.
- O'NIONS R. K., EVENSON N. M. and HAMILTON P. J. (1979) Geochemical modeling of mantle differentiation and crustal growth. *J. Geophys. Res.* **84**, 6091–6101.
- SACKETT W. M., POTRATZ H. A. and GOLDBERG E. D. (1958) Thorium content of ocean water. *Science* **128**, 204–205.
- SAFRONOV V. S. (1969) *Evolution of the Protoplanetary Cloud and Formation of the Earth and Planets*. Nauka, Moscow; translated for NASA and NSF by Israel Program for Scientific Translation (1972), NASA TT5-677.
- SUN S.-S. (1980) Lead isotopic study of young volcanic rocks from mid-ocean ridges, ocean islands and island arcs. *Phil. Trans. Roy. Soc. London* **297**, 409–445.
- TATSUMOTO M. (1978) Isotopic composition of lead in oceanic basalt and its implication to mantle evolution. *Earth Planet. Sci. Lett.* **38**, 63–87.
- TAYLOR S. R. (1987) The unique lunar composition and its bearing on the origin of the Moon. *Geochim. Cosmochim. Acta* **51**, 1297–1309.
- TAYLOR S. R. and MCLENNAN S. M. (1985) *The Continental Crust: Its Composition and Evolution*. Blackwell, 312p.
- TITAEVA N. A. (1984) Th-isotopic composition as indicator of volcanic rock origin (abstr.). *27th Intl. Geol. Cong. Abstr.*, Vol. V, 416–417.
- VIDAL P. and DOSSO L. (1978) Core formation: catastrophic or continuous? Sr and Pb isotope geochemistry constraints. *Geophys. Res. Lett.* **5**, 169–172.
- VOLLMER R. (1977) Terrestrial lead isotopic evolution and formation of the Earth's core. *Nature* **270**, 144–147.
- WETHERILL G. W. (1980) Formation of the terrestrial planets. *Ann. Rev. Astron. Astrophys.* **18**, 77–113.
- WHITE W. M. (1985) Sources of oceanic basalts: Radiogenic isotopic evidence. *Geology* **13**, 115–118.
- WHITE W. M. and PATCHETT J. C. (1984) Hf-Nd-Sr isotopes and incompatible element abundances in island arcs: Implications for magma origins and crust-mantle evolution. *Earth Planet. Sci. Lett.* **67**, 167–185.
- ZARTMAN R. E. (1974) Lead isotopic provinces in the cordillera of the Western United States and their geologic significance. *Econ. Geol.* **69**, 792–805.
- ZARTMAN R. E. (1984a) Lead, strontium, and neodymium isotopic characterization of mineral deposits relative to their geologic environments. In *Metallogenesis and Mineral Ore Deposits*, Vol. 12, Proc. 27th Intl. Geol. Cong., Moscow 4–14 August, 1984, pp. 83–106, VNU Science Press.
- ZARTMAN R. E. (1984b) The lower crust—missing reservoir in the

lead isotope paradox (abstr.). *Trans. Amer. Geophys. Union* **65**, 230.

ZARTMAN R. E. and DOE B. R. (1981) Plumbotectonics—The model. *Tectonophysics* **75**, 135–162.

ZINDLER A. and HART S. (1986) Chemical geodynamics. *Ann. Rev. Earth Sci.* **14**, 493–571.

APPENDIX: MATHEMATICAL FORMULATION

The following mathematical treatment of version IV of the plumbotectonics model closely resembles that of ZARTMAN and DOE (1981). However, because of the expanded size and greater complexity, a somewhat revised notational format has been adopted. With this new format it is possible to write single general equations for each of the several required operations. These general equations can then be applied to specific reservoirs and nuclides by assigning values to their various parameters. We proceed to derive the general equations for (1) mass transport between reservoirs, (2) isotopic species transport between reservoirs, including element fractionation, and (3) radioactive decay within reservoirs occurring between orogenies.

The mass transport equation between reservoirs at time, t , can be written as

$$\Delta M(t)^{\mu \rightarrow \nu} = M(t)^{\mu} \cdot \Phi(t)^{\mu \rightarrow \nu}$$

where $\Delta M(t)^{\mu \rightarrow \nu}$ is the increment of mass being transported from reservoir μ to reservoir ν , $M(t)^{\mu}$ is the mass of reservoir μ , and $\Phi(t)^{\mu \rightarrow \nu}$ is a parameter specifying the mass fraction of reservoir μ removed. As shown in Table A1, $\Phi(t)^{\mu \rightarrow \nu}$ can be constant or time dependent. After all permutations of $\Phi(t)^{\mu \rightarrow \nu}$ are numerically evaluated, the mass distribution among reservoirs is uniquely determined and can be calculated at the time of each orogeny.

Where $\Phi(t)^{\mu \rightarrow \nu}$ is functionally related to the magnitude of orogenic activity at time, t , the absolute values of this parameter can be defined by $\Sigma \Phi(t)^{\mu \rightarrow \nu} = \text{Constant}$, where the summation is made over all t . For example, in version IV increments of mass are transferred from the mantle to the distal component of the orogene in proportion to orogenic activity so that $\Sigma \Phi(t)^{\mu \rightarrow \nu} = 10$. That is, over the lifetime of the Earth the mantle is recycled ten times through the distal component of the orogene.

Table A1. Functions of gates (direction is with respect to orogene from reservoir μ to reservoir ν). Refer to Figure 1.

Gate	Direction	$\mu \rightarrow \nu$ (,v")	$\Phi(t)^{\mu \rightarrow \nu}$ (,v")	$\Delta \nu', \nu''$	Element bias
A1	Toward	m+r	Variable	--	No
A2	Toward	r+d	0.20	--	Yes
A3	Toward	$\Sigma v+d,p$	0.05	0.14	Yes
A4	Toward	$\Sigma u+p$	Variable	--	No
A5	Toward	$\Sigma l+p$	Variable	--	No
A6	Toward	$\Sigma s+w$	Variable	--	No
A7	Toward	m+w	Variable	--	No
B1	Within	d+p	Variable	--	Yes
B2	Within	d+w	Variable	--	Yes
B3	Within	w+p	n.d.	--	Yes
C1	Away	r+m	1.00	--	No
C1	Away	d+m	1.00	--	No
C2	Away	p+u,l	1.00	0.50	Yes
C3	Away	w+s	1.00	--	No

m - mantle, r - mid-ocean ridge, d - distal component of orogene, p - proximal component of orogene, w - mantle wedge component of orogene, u - upper crust, l - lower crust, s - subcrustal lithosphere, v - vertically recycled component of upper crust, n.d. - not defined for version IV.

Table A2. Distribution coefficients, $\alpha_{E^{\mu \rightarrow \nu}}$ and $\alpha_{F^{\mu \rightarrow \nu}, \nu''}$, for Pb, Th, U, Sr, Rb, Nd, and Sm.

Element	$E^{\mu \rightarrow d}$	$E^{\mu \rightarrow p}$	$E^{\mu \rightarrow w}$	$F^{\nu \rightarrow d,p}$	$F^{\mu \rightarrow u,l}$
Pb	80	40	25	1.00	3.67
Th	100	47	25	0.67	4.00
U	100	52	27	1.50	6.25
Sr	12	8	5	1.00	1.00
Rb	50	23	10	1.00	9.00
Nd	10	6	5	1.00	2.33
Sm	8	5	3	1.00	2.25

r - mid-ocean ridge, d - distal component of orogene, p - proximal component of orogene, w - mantle wedge component of orogene, u - upper crust, l - lower crust, v - vertically recycled component of upper crust.

The horizontal recycling function, which transports portions of preexisting upper crust and lower crust to the proximal component of the orogene, and subcrustal lithosphere to the mantle wedge component of the orogene, is also related to the magnitude of orogenic activity. In this case a summation is chosen that gives the correct area of modern continental crust, and, therefore, that depends on the rate of production of crust. In version IV, crustal area must be reduced by 33 percent in order to arrive at its final configuration (the upper crust is also subject to a vertical recycling function, so it experiences considerably more recycling).

In instances where the donor reservoir consists of separate segments, which individually contribute to the total increment of mass, a summation, designated $\Sigma \mu$, is carried out over these segments. Preexisting upper crust, lower crust, and subcrustal lithosphere—formed during each preceding orogeny—are comprised of such segments.

If an increment of mass is transported into two receiving reservoirs, ν' and ν'' , a branching function, $\Lambda^{\nu',\nu''}$ ($=X$ for reservoir ν' and $1-X$ for reservoir ν'' ; $0 \leq X \leq 1$), defines the fraction of mass going into each reservoir. In all cases where a branching function is used, that function is time-independent. Thus, the generalized mass transport equation becomes

$$\Delta M(t)^{\Sigma \mu \rightarrow \nu', \nu''} = M(t)^{\Sigma \mu} \cdot \Phi(t)^{\Sigma \mu \rightarrow \nu', \nu''} \cdot \Lambda^{\nu', \nu''}$$

Next let us consider the transport of increments of isotopic species between reservoirs. Similarly to the mass transport equation, we can write

$$\Delta^{\alpha} N(t)^{\mu \rightarrow \nu} = {}^{\alpha} N(t)^{\mu} \cdot {}^{\alpha} \theta(t)^{\mu \rightarrow \nu}$$

where $\Delta^{\alpha} N(t)^{\mu \rightarrow \nu}$ is the number of moles of species α being transported from reservoir μ to reservoir ν , ${}^{\alpha} N(t)^{\mu}$ is the number of moles of species α in reservoir μ , and ${}^{\alpha} \theta(t)^{\mu \rightarrow \nu}$ is a parameter specifying the mole fraction of species α removed from reservoir μ . In general, we can relate ${}^{\alpha} \theta(t)^{\mu \rightarrow \nu}$ to $\Phi(t)^{\mu \rightarrow \nu}$ as follows

$${}^{\alpha} \theta(t)^{\mu \rightarrow \nu} = \Phi(t)^{\mu \rightarrow \nu} \cdot {}^{\alpha} E^{\mu \rightarrow \nu} / [\Phi(t)^{\mu \rightarrow \nu} ({}^{\alpha} E^{\mu \rightarrow \nu} - 1) + 1]$$

or, rearranging

$${}^{\alpha} E^{\mu \rightarrow \nu} = [{}^{\alpha} \theta(t)^{\mu \rightarrow \nu} / (1 - {}^{\alpha} \theta(t)^{\mu \rightarrow \nu})] / [\Phi(t)^{\mu \rightarrow \nu} / (1 - \Phi(t)^{\mu \rightarrow \nu})]$$

where ${}^{\alpha} E^{\mu \rightarrow \nu}$ is the partitioning coefficient for species α between the removed increment and the remainder of reservoir μ . This definition of ${}^{\alpha} E^{\mu \rightarrow \nu}$ differs from its original usage in ZARTMAN and DOE (1981) as an enrichment factor over the initial concentration of reservoir μ . All isotope partitioning coefficients are considered to be time independent and assume the same value for species of the same element. Combining Eqns. (3) and (4), and expanding to generalized form, the isotope transport equation becomes

$$\Delta^{\alpha} N(t)^{\Sigma \mu \rightarrow \nu} = {}^{\alpha} N(t)^{\Sigma \mu} \cdot \Phi(t)^{\Sigma \mu \rightarrow \nu} \cdot {}^{\alpha} E^{\mu \rightarrow \nu} / [\Phi(t)^{\Sigma \mu \rightarrow \nu} ({}^{\alpha} E^{\mu \rightarrow \nu} - 1) + 1]$$

Where branching into two receiving reservoirs is involved, we also must define the consequent distribution of isotopic species. However,

in this case the partitioning coefficient, ${}^{\alpha}F^{\mu \rightarrow \nu', \nu''}$, controls the proportionment of the isotope between the two receiving reservoirs ν' and ν'' . Two such branchings are used in the model: (a) the vertically recycled component of the *upper crust* is divided between the *distal* and *proximal* components of the *orogene*, and (b) the *proximal* com-

ponent of the *orogene* disperses to form the *upper* and *lower crust*. With our notation, ${}^{\alpha}F^{\mu \rightarrow \nu', \nu''}$ gives the partitioning coefficient for reservoir ν' , and ${}^{\alpha}F^{\mu \rightarrow \nu'', \nu'} = 1/{}^{\alpha}F^{\mu \rightarrow \nu', \nu''}$. The number of moles of species α going into reservoir ν' is

$$\Delta^{\alpha}N(t)^{\mu \rightarrow \nu'} = \Delta^{\alpha}N(t)^{\Sigma \mu \rightarrow \nu', \nu''} \cdot \Lambda^{\nu', \nu''} \cdot {}^{\alpha}F^{\mu \rightarrow \nu', \nu'} / [\Lambda^{\nu', \nu''} ({}^{\alpha}F^{\mu \rightarrow \nu', \nu''} - 1) + 1]$$

and the number of moles of species α going into reservoir ν'' is

$$\Delta^{\alpha}N(t)^{\mu \rightarrow \nu''} = \Delta^{\alpha}N(t)^{\mu \rightarrow \nu', \nu''} - \Delta^{\alpha}N(t)^{\mu \rightarrow \nu'}$$

The preceding mass and isotope transport equations apply sequentially to operations toward, within, and away from the *orogene*. Each operation behaves as a 'gate', and a brief description of each gate's function keyed to Fig. 1 is given in Table A1. Partitioning coefficients, ${}^{\alpha}E^{\mu \rightarrow \nu}$ and ${}^{\alpha}F^{\mu \rightarrow \nu', \nu''}$, for the elements under investigation are given in Table A2. Starting conditions for version IV are given in Table A3. Numerical values of all model parameters, including the concentration and isotopic composition for each reservoir as a function of time, are retrievable from the HP Series 200/300 BASIC computer program (HAINES and ZARTMAN, 1988).

Finally, we consider the accumulation of radiogenic species from the decay of their radioactive parents during residence in reservoirs following one orogeny and preceding the next orogeny. Orogenies themselves are modelled as instantaneous events and we do not treat radioactive decay in an *orogene*. By convention, all equations are given in terms of parent isotope normalized to its equivalent modern abundance. The increase in abundance of daughter isotopes acquired between two successive orogenies occurring at times, t' (older) and t'' (younger), is

$${}^{\beta}N(t'')^{\mu} = {}^{\beta}N(t')^{\mu} + {}^{\alpha}N(t')^{\mu} \cdot (e^{\lambda t'} - e^{\lambda t''})$$

where species α is the radioactive parent, species β is its radiogenic daughter, and λ is the corresponding decay constant.

Table A3. Starting conditions for version IV with all material in undifferentiated upper mantle to 670 km depth (1050×10^{24} g) at 4.2 Ga.

A. Initial isotopic composition	
$^{206}\text{Pb}/^{204}\text{Pb}$	10.06
$^{207}\text{Pb}/^{204}\text{Pb}$	11.40
$^{208}\text{Pb}/^{204}\text{Pb}$	29.58
$^{87}\text{Sr}/^{86}\text{Sr}$	0.6994
$^{143}\text{Nd}/^{144}\text{Nd}$	0.50716
B. Element abundances	
^{204}Pb	19.0×10^{-15} moles
^{238}U	177×10^{-15} moles
^{232}Th	758×10^{-15} moles
^{86}Sr	22000×10^{-15} moles
^{87}Rb	2140×10^{-15} moles
^{144}Nd	2100×10^{-15} moles
^{147}Sm	426×10^{-15} moles